Schwarzschild And Kerr Metric: Physics Of Black Holes And Curvature Of SpaceTime

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1 Schwarzschild Metric

1.1 Historical Background

The Schwarzschild metric was created by the outstanding German astronomer and physicist Karl Schwarzschild in 1915, who laid the foundations of the mathematical apparatus of modern relativistic astrophysics. The Schwarzschild metric (also known as the Schwarzschild solution) is an exact solution to the Einstein field equations that describes the gravitational field outside a spherical mass, on the assumption that the electric charge of the mass, angular momentum of the mass, and universal cosmological constant are all zero. The solution is a useful approximation for describing slowly rotating astronomical objects such as many stars and planets, including Earth and the Sun. It was found by Karl Schwarzschild and independently of him by Johannes Droste in 1916.

Schwarzschild sent a manuscript detailing his exact solution to Einstein in late December 1915. Einstein was astounded and delighted that such a clean, closed-form solution could be found. He formally presented Schwarzschild's paper to the Prussian Academy of Sciences in January 1916.

1.2 Meaning of Schwarzschild Metric

The Schwarzschild metric describes time flows near a static black hole, because by Theory of General Relativity we know that time depends on gravity and position

1.3 The Mathematical Formulation

The line element is given by:

$$ds^2 = -\left(1 - \frac{2GM}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta \, d\phi^2$$

G is the gravitational constant
M is the central mass
c is the speed of light
r is the radial coordinate (areal radius)
t is the coordinate time (for a distant observer)
θ, ϕ are the usual spherical angles

1.4 Common Simplified Forms

Geometric units ($G = c = 1$):

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$. With Schwarzschild radius $r_s = \frac{2GM}{c^2}$:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

1.5 Key Features

Event horizon: At $r = r_s = 2GM/c^2$. This is a coordinate singularity in these coordinates (removable with other coordinate systems like Kruskal–Szekeres), but it marks the boundary of a black hole. Singularity: True curvature singularity at $r = 0$. Asymptotically flat: As $r \to \infty$, the metric approaches Minkowski spacetime (special relativity). Static and spherically symmetric. Perihelion precession of orbits and gravitational redshift / light bending are natural consequences. Reduces to Newtonian gravity in the weak-field, slow-motion limit.

1.6 Important Radii

Schwarzschild radius $r_s = 2GM/c^2$ (event horizon for a black hole). Photon sphere (unstable circular photon orbits) at $r = 1.5r_s = 3GM/c^2$. ISCO (Innermost Stable Circular Orbit) at $r = 3r_s = 6GM/c^2$ for massive particles.

2 Kerr Metric

2.1 Historical Background

Kerr metric was discovered by New Zealand mathematician and physicist Roy Patrick Kerr. After the 1916 discovery of the Schwarzschild metric (for non-rotating black holes), scientists struggled for 47 years to find a solution that included rotation, a necessary feature for real-world stars and galaxies. In 1963, New Zealand mathematician Roy Kerr found the solution. It was a shock to the physics community, as many had doubted an exact solution for a rotating body was even possible. Kerr used a clever algebraic approach rather than standard, brute-force calculations, uncovering a geometry far more complex than a non-rotating black hole. Since almost all massive objects in the universe rotate, the Kerr metric is the "gold standard" for modeling actual black holes. It predicted unique phenomena like frame-dragging (the rotation of the black hole actually "drags" the surrounding spacetime with it) and the ergosphere (a region outside the event horizon where space itself is forced to rotate).

2.2 Meaning of Kerr Metric

Kerr metric describes the measurement of time changes near a rotating black hole

2.3 The Mathematical Formulation

Standard Form: Boyer-Lindquist Coordinates

The most common expression for the line element (in units where $c = 1$, often with Schwarzschild radius $r_s = 2M$) is:

$$ds^2 = -\left(1 - \frac{r_s r}{\Sigma}\right) dt^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \left(r^2 + a^2 + \frac{r_s r a^2}{\Sigma} \sin^2\theta\right) \sin^2\theta \, d\phi^2 - \frac{2 r_s r a \sin^2\theta}{\Sigma} \, dt \, d\phi \tag{1}$$

Where:

$\Sigma = r^2 + a^2 \cos^2\theta$

$\Delta = r^2 - r_s r + a^2$

(Other forms include Kerr-Schild coordinates or ingoing Eddington-Finkelstein coordinates.)

2.4 Common Simplified Forms

1. Boyer-Lindquist Coordinates (Most Common)

This is the standard "Schwarzschild-like" form. In geometric units ($G = c = 1$), with $r_s = 2M$:

$$ds^2 = -\left(1 - \frac{2Mr}{\Sigma}\right) dt^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma \, d\theta^2 + \left(r^2 + a^2 + \frac{2Mra^2}{\Sigma} \sin^2\theta\right) \sin^2\theta \, d\phi^2 - \frac{4Mra \sin^2\theta}{\Sigma} \, dt \, d\phi \tag{2}$$

Key functions:

$$\Sigma = r^2 + a^2 \cos^2\theta \tag{3}$$

$$\Delta = r^2 - 2Mr + a^2 \tag{4}$$

This form clearly shows the cross-term $dt \, d\phi$ responsible for frame-dragging. When $a = 0$, it reduces exactly to the Schwarzschild metric.

2. Kerr-Schild Coordinates (Elegant and Useful)

This form expresses the metric as flat Minkowski spacetime plus a simple rank-2 correction:

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu + 2H k_\mu k_\nu dx^\mu dx^\nu \tag{5}$$

(or equivalently $g_{\mu\nu} = \eta_{\mu\nu} + f k_\mu k_\nu$).
In Cartesian-like coordinates (t, x, y, z):

$$H = \frac{Mr^3}{r^4 + a^2 z^2} \tag{6}$$

(one common convention uses $f = \frac{2Mr^3}{r^4+a^2z^2}$).
Explicit form (one common version):

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 + \frac{2Mr^3}{r^4 + a^2 z^2} \left(dt + \frac{r(x\,dx + y\,dy) - a(y\,dx - x\,dy) + z\,dz}{r^2 + a^2} \right)^2 \tag{7}$$

Advantages:

- Only one extra term beyond flat space.
- The determinant is -1 everywhere (a very nice property).
- Excellent for perturbations, numerical relativity, and exact solutions.

3 Comparison Of Two Metrics

Let's start by comparing how Schwarzschild and Kerr black holes bend spacetime.

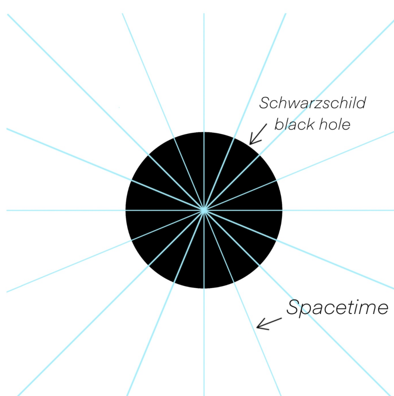


Figure 1: Here's how curvature Schwarzschild's black hole

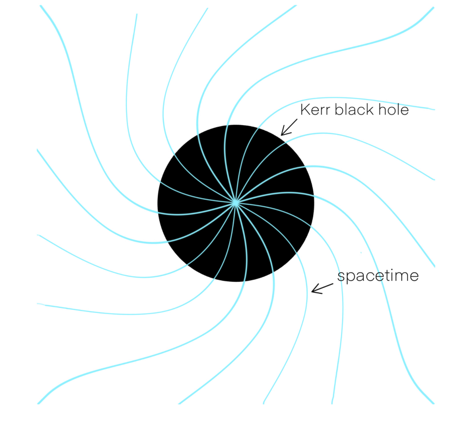


Figure 2: And This is how the curvature of spacetime changes due to the rotation of the Kerr black hole

4 Visual Differences

One of the visual differences is the shape of the Schwarzschild and Kerr black holes

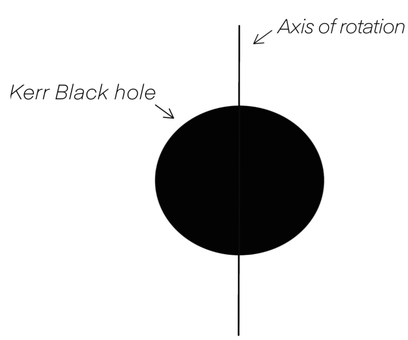
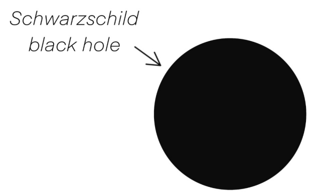


Figure 3: The Kerr black hole is not a perfect sphere, unlike the Schwarzschild black hole.



4.1 The most noticeable visual difference

If an observer theoretically approaches both black holes, the most noticeable difference will be the glow of the accretion disk orbiting the Kerr black hole. This happens because the rotation of the Kerr black hole causes one side of the accretion disk to move towards us. This glow is brighter on one side and darker on the other due to the black hole's rotation. This phenomenon is called the Doppler Effect.

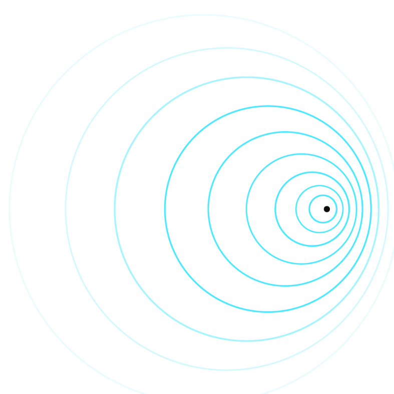


Figure 4: visualisation of doppler effect

5 Conclusion

The Schwarzschild and Kerr metrics represent the cornerstones of our understanding of black holes, illustrating how gravity shapes the very fabric of our universe. While the Schwarzschild metric provides a pristine, static model, the Kerr metric introduces the complexity of rotation, revealing the fascinating phenomena of frame-dragging and ergospheres. Together, these solutions do more than just describe gravitational singularities—they challenge our perception of space and time, bridging the gap between theoretical equations and the observable reality of the cosmos. As we continue to explore the mysteries of quantum gravity, these foundational models remain the essential ”gold standard” that guides our journey into the unknown.